

EDUCATION

Indian Institute of Technology(IIT), Bhubaneswar	September 2020 - May 2025
B.Tech + M.Tech (Dual Degree) in Electrical Engineering	Current CGPA: 7.06 / 10

EXPERIENCE

Indian Institute of Technology, Bhubaneswar	July 2024 - May 2025
M.Tech Research Fellow(ML, Deep Learning)	

- Developed new fault-detection models using Neural Networks for buck converters and 3-phase inverters to improve system reliability
- Optimized model performance for accurate fault classification under varying operating conditions
- Debugged complex issues in model predictions and improved fault detection robustness

Indian Institute Of Technology, Bhubaneswar	May 2024 - July 2024
Research Intern(ML)	

- Developed a machine learning-based system for detecting gate driver faults in EV chargers using simulated grid, converter, and control unit data
- Built reusable datasets for fault scenario generation and testing Optimized system accuracy (95%) and performance across different fault conditions
- Implemented and debugged real-time monitoring and alert mechanisms for fault detection Worked for handling simulation outputs and model inputs

PROJECTS

Genius: AI Powered SaaS Application 	Web Dev, AI
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- Developed new user-facing features using React.js and Next.js for AI-driven multimedia content generation.
- Built reusable components and front-end libraries using TypeScript for scalable UI development.
- Integrated RESTful APIs (Gemini API, Replicate API) for AI functionality Implemented modern authorization mechanisms using Clerk and subscription workflows via Stripe.
- Used PostgreSQL as the backend database to efficiently store, manage, and retrieve application data.

SHREX: Hotel Booking WebApp (MongoDb, ReactJs, ExpressJs, Razorpay API) 	Web Dev
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- Developed new user-facing features using React.js for hotel booking and dashboard interfaces.
- Built reusable UI components for booking flows, filters, and admin panels.
- Integrated RESTful APIs for booking management and Razorpay payment gateway.
- Optimized front-end performance using local storage, efficient state handling.

Client 2 Client Connection using TCP sockets with Encryption 	C++
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- Implemented TCP server client interaction which is a client server based chat application. Encrypted the chats using RC4 Encryption and used DIFFIE-HELLMANN algorithm for Key Exchange.
- Engineered a server-client architecture, enabling clients to view a list of connected clients and their status(BUSY,FREE etc), facilitating match requests and establishing separate communication channels based on client availability.

TECHNICAL SKILLS

- **Languages:** C++, JavaScript, TypeScript, Python, MySQL, MS Office
- **Web Development:** HTML5, CSS3, Tailwind CSS, ReactJS, NextJS, NodeJS, ExpressJS
- **Database:** NoSQL-MongoDB — SQL-MySQL
- **Others:** Git, GitHub, Kaggle, Jupyter, LaTeX, Postman, CAD

CODING PROFILE

- **LeetCode:** https://leetcode.com/u/Dev-011_m/
- **GeeksForGeeks:** <https://www.geeksforgeeks.org/user/devmorwal261/>

RELEVANT COURSEWORK

- **Computer Science :** Data Structures and Algorithms (DSA), Object-Oriented Programming System (OOPs), Software Development Life Cycle (SDLC), Operating System (OS).
- **Mathematics:** Probability and Combinatorics, Number Theory, Set Theory

EXTRA CURRICULAR ACTIVITIES

- Core Head of Alma Fiesta (The Socio-Cultural Fest), IIT Bhubaneswar.
- Led photography team at Inter IIT Cult Meet 2023.
- Clix, Photography Society, IIT Bhubaneswar.
- Secured 2nd place in Badminton at the General Championship.